Chapter 10, Problem 51P

Problem

Use source transformation to find **I***o* in the circuit of Prob. 10.42.

Step-by-step solution

Step 1 of 7

Given circuit is,

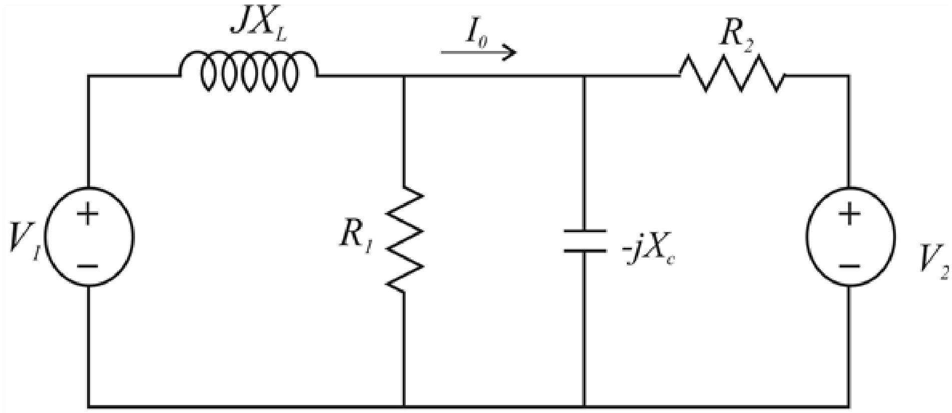


Figure-(1)

Step 2 of 7

We transform the voltage source to current source, is shown below

$I_1 = \frac{V_1}{jX_L}$ and

$I_2 = \frac{V_2}{P_2}$

Step 3 of 7

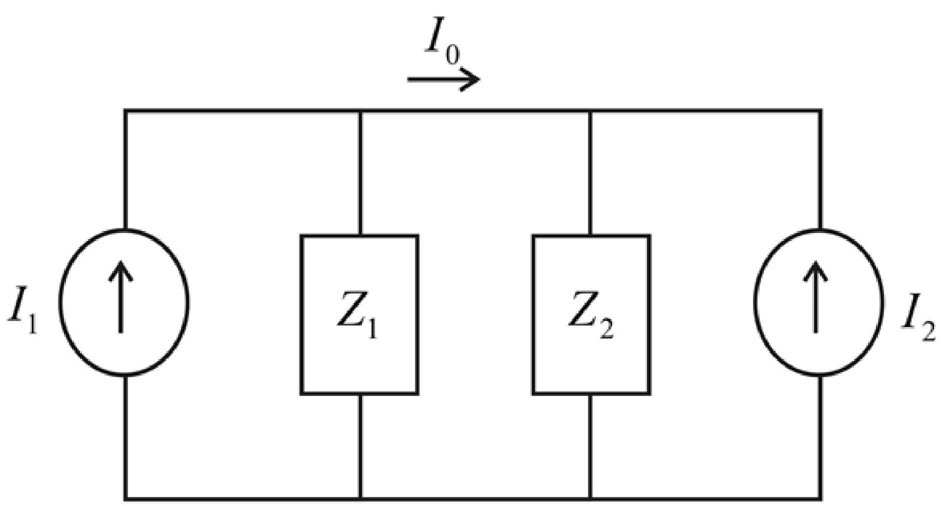


Figure-(2)

Step 4 of 7

Where,

$$Z_1 = jX_L \parallel R_1$$

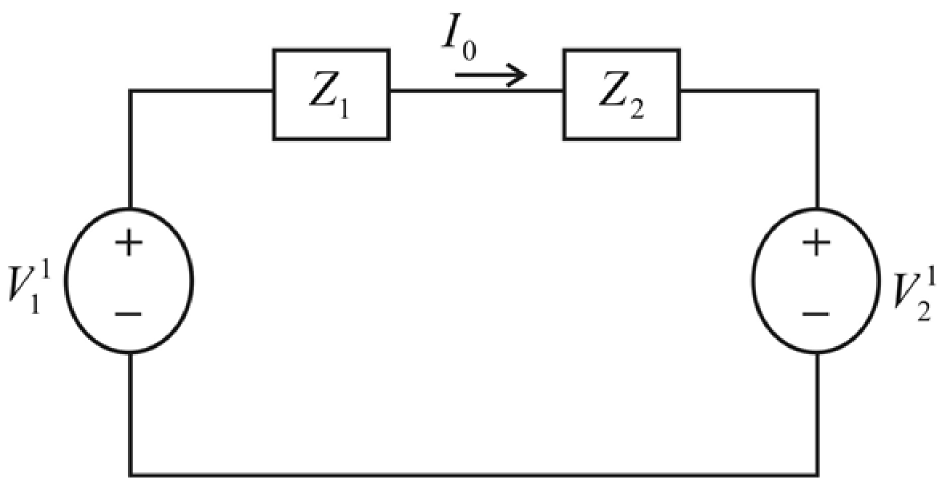
$$Z_1 = \frac{jX_L R_1}{R_1 + jX_L}$$

$$Z_2 = -jX_C \parallel R_2$$

$$Z_2 = \frac{-jX_C R_2}{R_2 - jX_C}$$

We transform the current source to voltage source, is shown below.

Step 5 of 7



Step 6 of 7

$$V_1^1 = I_1 Z_1$$

$$V_2^1 = I_2 Z_2$$

$$V_1^1 = \frac{V_1}{jX_L}\left[\frac{jX_L R_1}{R_1 + jX_L}\right]$$

$$V_2^1 = \frac{V_2}{R_2}\left[\frac{-jX_c R_2}{R_2 + jX_C}\right]$$

Applying *KVL* above circuit:

$$-V_1^1 + Z_1 I_0 + Z_2 I_0 + V_2^1 = 0$$

$$I_0\left(Z_1 + Z_2\right) = V_1^1 - V_2^1$$

$$I_0 = \frac{V_1^1 - V_2^1}{Z_1 + Z_2}$$

Step 7 of 7

$$I_0 = \frac{\dfrac{V_1}{jX_L}\left[\dfrac{jX_L R_1}{R_1 + jX_L}\right] - \dfrac{V_2}{R_2}\left[\dfrac{-jX_C R_2}{R_2 - jX_C}\right]}{\left[\dfrac{jX_L R_1}{R_1 + jX_C}\right] + \left[\dfrac{-jX_C R_2}{R_2 - jX_C}\right]}$$

$$I_0 = \frac{V_1\left[\dfrac{R_1}{R_1 + jX_L}\right] - V_2\left[\dfrac{-jX_C}{R_2 - jX_C}\right]}{\left[\dfrac{jX_L R_1}{R_1 + jX_C}\right] + \left[\dfrac{-jX_c R_2}{R_2 - jX_C}\right]}$$

$$I_0 = \frac{V_1 R_1\left(R_2 - jX_C\right) + jV_2 X_C\left(R_1 + jX_L\right)}{jX_L R_1\left(R_2 - jX_C\right) - jX_C R_2\left(R_1 + jX_L\right)}$$

$$I_0 = \frac{\left(V_1 R_1 R_2 - V_2 X_L X_c\right) + j\left(V_2 R_1 X_C - V_1 R_1 X_c\right)}{\left(R_1 X_C X_L + R_2 X_C X_L\right) + j\left(X_L R_1 R_2 - X_C R_1 R_2\right)}$$

$$\boxed{I_0 = \frac{\left(V_1 R_1 R_2 - V_2 X_L X_c\right) + jR_1 X_c\left(V_2 - V_1\right)}{X_c X_L\left(R_1 + R_2\right) + jR_1 R_2\left(X_L - X_c\right)}}$$